eureka-digits-sol: reproducible research report

Libor “Burny” Burian

Abstract

This report records a bounded research cycle executed through Codex. The mean paired accuracy difference is 0.000000, with conditional 95% bootstrap interval [0.000000, 0.000000]. The registered decision is **inconclusive**. The sample standard deviation across seed effects is 0.000000. The interval uses 10000 paired resamples of test examples and conditions on the trained models.

Recorded harness status: **complete**; 8 operational phases accepted. Measurements below do not imply completion of unaccepted phases.

Context, motivation, and introduction

Does AdamW weight decay improve held-out accuracy of a small MLP on the bundled digits data?

The objective is reproducible evidence and an inspectable research process. This study does not establish broad scientific reliability or superiority of an agent model.

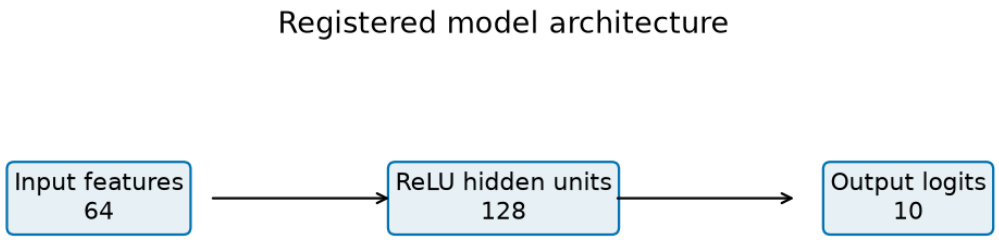
Background and method

- **analysis_kind**: “exploratory”
- **prior_results_known**: true
- **objective**: “Replicate a known fixed study to evaluate engineering completion; do not claim new scientific confirmation.”
- **question**: “Does AdamW weight decay improve held-out accuracy of a small MLP on the bundled digits data?”
- **architecture**: [64, 128, 10]
- **optimizer**: “AdamW”
- **learning_rate**: 0.001
- **batch_size**: 64
- **epochs**: 50
- **methods**: [“regularized”, “unregularized”, “sensitivity”, “shuffled_labels”, “logistic”, “class_prior”]
- **seeds**: [0, 1, 2]
- **data_boundaries**: “Frozen stratified 60/20/20 train/validation/test partition. Split seed 42. Divide pixels by the documented constant 16. Pilot reports validation only. No learned preprocessing or selection uses test data. Generalization is limited to this random example split, not new writers or datasets. Prior results on this evaluation set are known. All replication observations and conclusions are exploratory; no untouched-data claim is made.”
- **analysis_plan**: “Use 10000 paired bootstrap resamples of test example indices with RNG seed 2026, preserving pairing across methods and all trained seeds. This interval conditions on the trained models and does not estimate training-population uncertainty. Separately show each seed effect and its sample standard deviation. Treat other comparisons as descriptive, with no post-hoc primary selection.”
- **hyperparameter_rationale**: “Small prespecified network and fixed training budget provide fast feedback on an established task. The three decay values are a descriptive local sensitivity check, not a

test-selected sweep. Three seeds expose limited initialization sensitivity; they do not support broad population claims."



Research architecture.

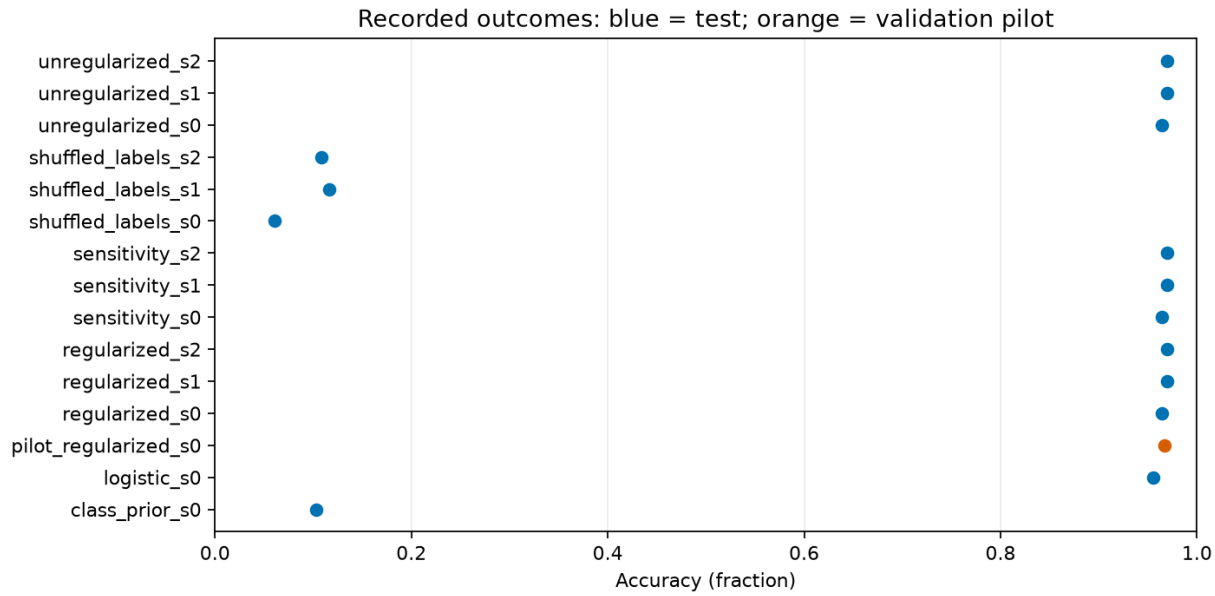


Registered model architecture.

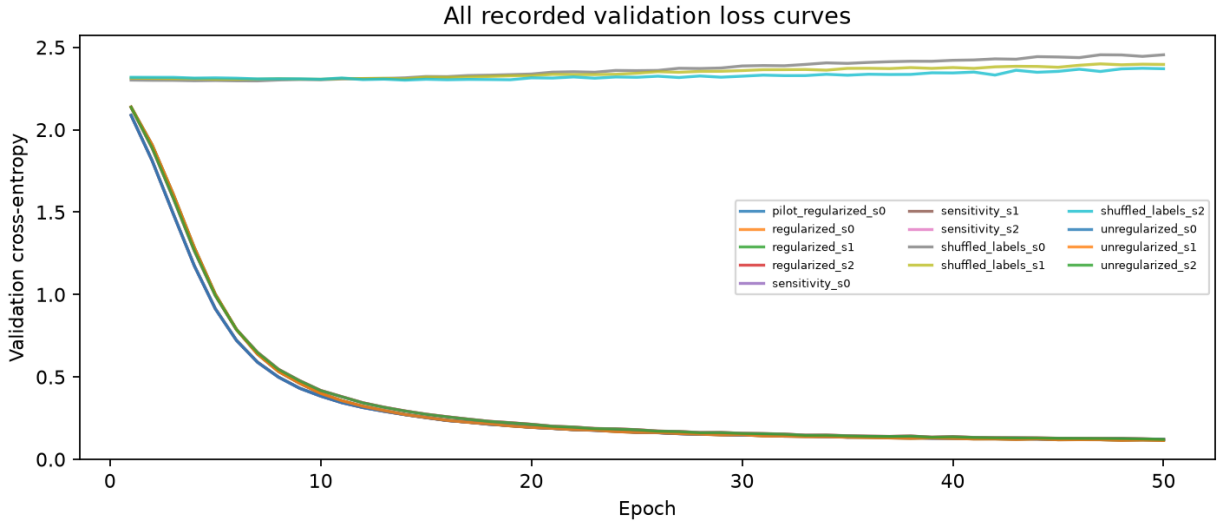
Results

Method	Seed	Split	Accuracy	Seconds	Device
class prior	0	test	0.102778	0.027	cpu
logistic	0	test	0.955556	0.109	cpu
regularized	0	validation	0.966574	6.083	cuda
regularized	0	test	0.963889	0.373	cuda
regularized	1	test	0.969444	5.740	cuda
regularized	2	test	0.969444	2.151	cuda
sensitivity	0	test	0.963889	2.098	cuda
sensitivity	1	test	0.969444	2.105	cuda
sensitivity	2	test	0.969444	2.121	cuda
shuffled labels	0	test	0.061111	2.190	cuda
shuffled labels	1	test	0.116667	2.194	cuda
shuffled labels	2	test	0.108333	2.103	cuda
unregularized	0	test	0.963889	2.032	cuda
unregularized	1	test	0.969444	2.142	cuda
unregularized	2	test	0.969444	2.203	cuda

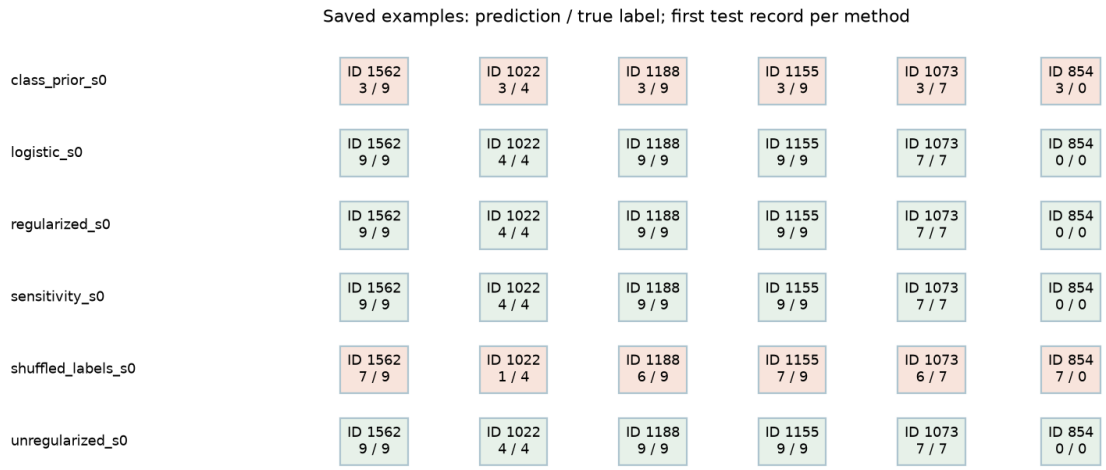
The mean paired accuracy difference is 0.000000, with conditional 95% bootstrap interval [0.000000, 0.000000]. The registered decision is **inconclusive**. The sample standard deviation across seed effects is 0.000000. The interval uses 10000 paired resamples of test examples and conditions on the trained models.



All recorded accuracies.



Recorded learning curves.



First saved test examples; no correctness-based selection.

Claims and evidence

- **claim-decay-tie**: In every registered paired fitted instance, the primary decay intervention and zero-decay ablation have identical test correctness vectors despite different saved model parameters. The registered conditional example-bootstrap analysis is inconclusive: the predicted positive accuracy benefit was not observed. Evidence: regularized_s0, regularized_s1, regularized_s2, unregularized_s0, unregularized_s1, unregularized_s2, audit-experiments-artifacts, src-bootstrap. Limitations: ['This is a decay intervention within AdamW, not a comparison between optimizers or with an L2 loss penalty.', 'The degenerate interval follows from identical observed correctness differences; it does not establish equivalence, a universal null, or uncertainty over future training runs.', 'Prior outcomes were known. The fixed example split does not establish generalization to new writers or datasets; probabilities and behavior outside the scored examples were not compared.']
- **claim-local-sensitivity**: The registered weaker-decay sensitivity arm has the same test accuracy as both primary arms within each registered seed. This local coefficient check found no accuracy advantage among the supplied decay settings. Evidence: sensitivity_s0, sensitivity_s1, sensitivity_s2, regularized_s0, regularized_s1, regularized_s2, unregularized_s0, unregularized_s1, unregularized_s2. Limitations: ['Equal accuracy does not imply equal individual predictions, confidence, calibration or behavior on other examples.', 'The supplied coefficient sensitivity is descriptive, not a search for an optimum or a test of architecture, learning rate, training duration or split sensitivity.']
- **claim-controls-and-baseline**: Every real-label neural result and the penalized logistic baseline exceed every shuffled-label result and the constant training-majority baseline on the fixed test split. Logistic accuracy is descriptively below every real-label neural result, while providing a useful simple baseline. Evidence: regularized_s0, regularized_s1, regularized_s2, unregularized_s0, unregularized_s1, unregularized_s2, sensitivity_s0, sensitivity_s1, sensitivity_s2, shuffled_labels_s0, shuffled_labels_s1, shuffled_labels_s2, logistic_s0, class_prior_s0, src-logistic. Limitations: ['These are descriptive fitted-instance comparisons; no registered inferential superiority or theoretical chance test was performed for these contrasts.', 'Poor shuffled-label generalization narrows some gross label or scoring failures but cannot certify absence of leakage, related examples or writer dependence.', 'The majority predictor is deterministic, not random class sampling; the logistic model is regularized, not the zero-decay neural ablation.']
- **claim-local-artifact-usability**: The full supplied comparison matrix has saved predictions and locally usable checkpoints that passed scoped split, label, arithmetic, provenance, history/log and reload checks. The matching final regularized arm reused the controller-published pilot checkpoint. The analysis consistency audit reproduced the existing audit and registered primary summary without fitting or rendering. Evidence: pilot_regularized_s0, audit-pilot-artifacts, audit-experiments-artifacts,

audit-analysis-consistency, src-reproducibility. Limitations: ['Passing these checks is evidence for this local execution, not a proof against all bugs or a claim of model, provider or harness superiority.', 'Reload verification does not establish cross-platform bitwise reproducibility, exact interrupted-training resumption or independent reproduction from a fresh environment.', 'Pilot scoring used validation labels, but reload verification forwarded test features. Controller publication and later review are separate from these local checks.']

Discussion, implications, and learnings

Interpretation of the fixed digits replication

Author: Libor "Burny" Burian. Stage: distillation. Analysis kind: **exploratory**. The engineering objective is verified completion with reproducible artifacts regardless of scientific effect direction. Known prior outcomes prevent new confirmation on untouched data. This report interprets preserved measurements; no training, test-selected tuning or new predictive evaluation was performed in analysis.

Motivation, context and claims

The scientific question is whether the supplied AdamW decay intervention improves held-out accuracy of the supplied small digit classifier. The strongest observation is narrower: the primary paired models have identical test correctness despite different fitted weights. The inherited rule calls the conditional result inconclusive. The positive directional prediction was not observed, but neither an equivalence claim nor a general rejection of weight decay follows. **claim-decay-tie**; evidence: regularized_s0, regularized_s1, regularized_s2, unregularized_s0, unregularized_s1, unregularized_s2, audit-experiments-artifacts.

The weaker-decay arm also ties the primary arms' per-seed accuracy. Real-label models exceed the negative controls, and penalized logistic regression supplies a useful simple baseline below the neural results descriptively. The saved artifacts pass the scoped local audits. These are the remaining claims in claims.json: **claim-local-sensitivity**, **claim-controls-and-baseline**, **claim-local-artifact-usability**. None establishes agent/model superiority or general scientific reliability.

All empirical scores, paired effects, interval endpoints, seed dispersion and parameter differences are generated from saved files in experiments.md. This interpretation deliberately links that table rather than hand-enter empirical scores. Raw experiment IDs resolve in results/; source IDs resolve in sources.json; local audit IDs resolve in evidence.json. The derived comparison-summary.json is not an additional experiment.

Methods and causal scope

The unchanged fixture and registered protocol define the full configuration. The prospective method table specifies architecture, all coefficients, learning rate, batch size, duration, seeds and logistic settings. These were inherited for inexpensive fixed feedback, not chosen from outcomes. The classifier is a fully connected hidden-ReLU MLP trained with cross-entropy and AdamW. Both primary arms use that optimizer; their matched initialization and minibatch-order construction isolate the decay coefficient within each seed. Decay applies to biases as well as weights. The weaker coefficient is a descriptive sensitivity arm, and training-label permutation is the negative-control intervention. Logistic regression is penalized and linear; the majority baseline predicts a constant class from training counts. Evidence: audit-fixture-source; src-torch-adamw, src-logistic.

Bundled digits are scaled by a known constant and partitioned by frozen, class-stratified example indices. No learned preprocessing statistic is estimated from held-out data. Parameters use training labels; validation loss is diagnostic and does not select an epoch or configuration. The pilot scores validation labels. Its reload check forwards all feature rows, including test inputs, so complete test-feature isolation cannot be claimed. Source inspection finds no test-label fitting or selection path; the artifact audit checks the saved membership, labels and scores. This is evidence for these specific boundaries, not a blanket certification of no leakage. Evidence: audit-fixture-source, audit-pilot-artifacts, audit-experiments-artifacts; src-digits-loader, src-splits.

The final fixed epoch supplies the neural checkpoints. The matching regularized test arm reuses the published pilot checkpoint rather than a new fit; the validation and test records are different evaluations of that fitted model and must not be counted as independent training runs. Evidence: pilot_regularized_s0, regularized_s0, audit-experiments-artifacts; controller record pilot-publication.json.

For seed set S and shared test examples T, define d_si as regularized correctness minus zero-decay correctness on example i. The primary statistic is the mean of d_si across seeds and examples. The analysis implementation draws test-example indices with replacement, shares those indices across methods and fitted seeds, and takes the registered central percentile endpoints. The frozen analysis plan fixes resample count, generator seed and quantile interpolation. Seed-effect sample dispersion is reported separately. This interval conditions on the fitted seed set; it does not resample training or estimate uncertainty across new writers or datasets. src-bootstrap explains shared-index pairing and distinguishes percentile intervals from the library's default BCa method; this study uses its explicit registered NumPy calculation.

Complete frozen settings

Generated directly from the registered protocol, without outcome-based selection.

Setting	Registered value
architecture	[64, 128, 10]
activation	"ReLU"
optimizer	"AdamW"
learning_rate	0.001
batch_size	64
epochs	50
seeds	[0, 1, 2]
split_seed	42
logistic_C	1.0
logistic_max_iter	1000
bootstrap_seed	2026
bootstrap_resamples	10000
weight_decay.regularized	0.01
weight_decay.unregularized	0.0
weight_decay.sensitivity	0.001
weight_decay.shuffled_labels	0.01

Predictions, observations and negative findings

Scientific predictions were recorded before fitting in preregistration.md and experiments-plan.md. Their prospective local timing does not remove prior-result exposure. The analysis audit expectations were separately recorded after results were known in analysis-plan.md; they are consistency predictions, not confirmatory hypotheses.

Prediction or alternative: The inherited h1 predicts a positive mean decay effect; no effect, harm and seed-dependent changes compete.

Observation: Every primary paired correctness vector matches. The saved aggregate and all resampled effects reproduce the degenerate tie.

Interpretation: The positive prediction was not observed. The interval is not wholly negative, so the registered directional falsification criterion is not met; the rule labels it inconclusive. Finite-example accuracy insensitivity remains compatible with changes in confidence or behavior elsewhere. Evidence: primary result IDs above; audit-analysis-consistency.

Prediction or alternative: Accidental reuse of identical primary checkpoints could explain a tie.

Observation: Paired state dictionaries differ in every primary pair.

Interpretation: This specific reuse explanation is inconsistent with the audited bytes. Different states alone do not demonstrate a regularization mechanism or calibration benefit. Evidence: audit-experiments-artifacts, reproduced by audit-analysis-consistency.

Prediction or alternative: The weaker coefficient might change accuracy; training seeds might matter more than the coefficient.

Observation: The sensitivity arm ties primary accuracy within each seed, while neural accuracy varies between registered seeds.

Interpretation: This narrow coefficient check found no accuracy advantage. It does not identify an optimum or establish stability over the training population. Evidence: sensitivity_s0, sensitivity_s1, sensitivity_s2 and primary result IDs.

Prediction or alternative: Real labels should outperform shuffled labels and the majority floor.

Observation: Every real-label neural result and logistic result exceed every negative-control result. Shuffled-label online training loss decreases while true-label validation loss ends above its initial value.

Interpretation: Compatible with learning useful label associations in the real-label arms and limited fitting of arbitrary labels in the control. It narrows gross label/scoring-error alternatives without excluding contamination or related-example dependence. Evidence: shuffled_labels_s0, shuffled_labels_s1, shuffled_labels_s2, class_prior_s0, logistic_s0 and real-label neural IDs.

Prediction or alternative: A simple linear baseline may be competitive.

Observation: Logistic accuracy is descriptively below each real-label neural result; generated scores show the size of the gap.

Interpretation: A useful linear baseline prevents attributing all useful accuracy to the neural architecture. No registered neural-versus-linear superiority interval or chance test was performed. Evidence: logistic_s0 and real-label neural IDs.

Prediction or alternative: Hashes, split membership, scores, histories and reloads should remain consistent.

Observation: The existing full artifact audit and primary summary reproduce; protected scientific files, results and checkpoints remain unchanged.

Interpretation: No checked provenance, score or reload defect was found. This does not prove every implementation property or general harness reliability. Evidence: audit-analysis-consistency.

The tie is a genuine negative finding for the predicted improvement in the observed statistic. If every d_si vanishes, any shared-example resample also has a vanishing mean. The interval's collapse therefore follows algebraically from these observed differences; it is not evidence of broad experimental power. Correctness equality also leaves open different wrong-class predictions. No additional probability, calibration, subgroup or new-data analysis was performed. Evidence: primary result IDs, audit-experiments-artifacts; analysis definition above.

Related work and implications

Loshchilov and Hutter distinguish decoupled decay from an L2 gradient penalty in adaptive optimization and report image-classification benefits. That motivates the component being replicated but does not predict an accuracy gain for this fixed digits setting. This experiment cannot reproduce the paper's optimizer comparison because it intervenes only on decay within AdamW. The original implementation and license were inspected during source research; no external implementation was copied here. src-adamw-paper, src-adamw-code.

The loader uses a subset of UCI optical digits, not MNIST. Repartitioning that subset randomly by example does not preserve the original dataset's writer-held-out construction; the current checks do not establish writer isolation. src-digits-loader, src-uci-digits. Meaningful simple baselines and explicit evaluation boundaries

follow the methodological concerns in src-lones-pitfalls; that guide is context, not empirical evidence for the local effect.

The closest supplied-fixture executions are src-prior-haiku and src-prior-luna. They are prior exposure, not local measurements or an agent ranking. Their Haiku disclosure (src-haiku-disclosure) clarifies pilot feature access; the Luna disclosure (src-luna-disclosure) motivates retaining historical bytes through presentation repairs. No sibling project was inspected.

Limitations, validity review and decision

The fixed small task, limited seeds, previously exposed outcomes and example-based partition restrict inference. Conditional example resampling assumes examples are appropriate resampling units; unmeasured writer or related-example dependence can weaken its sampling interpretation. No equivalence margin, untouched evaluation set, writer-held-out replication, split sensitivity or broader hyperparameter sweep was tested. A constant paired effect across these fitted seeds does not imply no training-population variability. Saved argmax predictions and loss histories do not establish confidence calibration, causal feature mechanisms or out-of-distribution behavior.

The validity review distinguishes source inspection, intervention evidence and post-hoc auditing. Matched decay interventions support the narrow local contrast; the label intervention supplies a negative control. The parameter comparison is a post-hoc artifact check, not a mechanistic demonstration. Recomputed labels, indices, hashes, metrics, log/history agreement and checkpoint outputs narrow specified ordinary-bug alternatives. Exact reloads show local artifact usability; optimizer state was not saved for exact interrupted-training resumption, and independent fresh-environment reproduction was not performed. Cross-platform identity is not guaranteed. Evidence: audit-fixture-source, audit-experiments-artifacts, audit-analysis-consistency; src-reproducibility.

Timestamped fitting logs and full negative-control results remain preserved. The expected test-first import failures documented in the core and comparison reports remain part of the engineering history; no scientific failed fit or discarded seed was found in the scoped log audit. The current non-training fixture check and focused statistic tests passed; their actual outputs are recorded in analysis-audit.json. No scientific definition, result record or checkpoint was edited, and no training or rendering ran in analysis. Historical controller-generated global summaries can still describe earlier accepted cutoffs; their refresh, final paper/site generation, publication and acceptance belong to the controller and later phases.

Writing exposed no substantive error requiring a return to training or protocol amendment. It sharpened the distinction between changed weights, unchanged correctness and conditional uncertainty. The useful next action is controller review of these bounded claims and the requirements map, preserving all negative findings. Further fitting after observing the test tie would answer a new question and has low information value for the authorized handoff. This moves from exploration through understanding to distillation, with a return justified only by a concrete evidence gap. src-nanda-process.

Limitations

- Conditional test-example intervals do not estimate variation over new training seeds, writers, or datasets.
- A small single-task study does not establish general model or harness superiority.
- No discordant predictions were observed in the primary pairs. The empirical bootstrap interval collapses to zero; this is not an equivalence test or a guarantee about unseen examples.

Exploratory outcomes do not become confirmatory by inclusion in this document. Missing metrics are not zero. Measurements without an applicable recorded objective are omitted; see the requirement checklist for explanations.

Related work and sources

- src-matplotlib-plot: Matplotlib Axes.plot API

- src-python-unittest: Python 3.14 unittest: discovery, assertions and isolated fixtures
- src-python-subprocess: Python 3.14 subprocess: argv sequences, pipes and process status
- src-adamw-paper: Decoupled Weight Decay Regularization — Loshchilov and Hutter, ICLR 2019
- src-adamw-code: AdamW-and-SGDW original implementation and license
- src-torch-adamw: PyTorch AdamW API and algorithm, version 2.14
- src-digits-loader: scikit-learn load_digits API
- src-uci-digits: Optical Recognition of Handwritten Digits — UCI, Alpaydin and Kaynak
- src-splits: scikit-learn train_test_split API
- src-logistic: scikit-learn LogisticRegression API
- src-bootstrap: SciPy bootstrap API: paired resampling and interval definitions
- src-reproducibility: PyTorch reproducibility notes, version 2.14
- src-nanda-process: How I Think About My Research Process: Explore, Understand, Distill — Neel Nanda
- src-lones-pitfalls: How to avoid machine learning pitfalls: a guide for academic researchers — Michael A. Lones
- src-prior-haiku: eureka-digits-haiku prior supplied-fixture execution
- src-prior-luna: eureka-digits-luna prior supplied-fixture execution
- src-haiku-disclosure: Haiku historical cutoff and subsequent publication disclosure
- src-luna-disclosure: Luna post-publication provenance audit disclosure
- src-pandoc: Pandoc User Guide: PDF creation, Markdown extensions and resource paths
- src-node-vm: Node.js vm API

Conclusion

The mean paired accuracy difference is 0.000000, with conditional 95% bootstrap interval [0.000000, 0.000000]. The registered decision is **inconclusive**. The sample standard deviation across seed effects is 0.000000. The interval uses 10000 paired resamples of test examples and conditions on the trained models.

Appendix: reproducibility and provenance

The committed protocol, per-experiment JSON files, saved predictions, loss histories, and checkpoints are the source of all numerical tables and figures. Rebuild using `eureka render`. See `docs/reproduction.md` for execution commands and environment details.

AI statement

Codex wrote the harness and study fixture; Codex performed this cycle and drafted its interpretation. Any subsequent parent corrections are documented in the linked audit reports. The harness generated this paper from recorded artifacts. A human supplied the exact specification. Human cleanup and rewriting are in progress.